

Spirit in Physics: Word-association latency and skin-potential response profiles in a five-participant pilot study

Abstract

We evaluated the feasibility of constructing participant-specific word-association response profiles from latency and skin-potential recordings and examined the repeatability of response latency. Eleven participant directories were present; five provided 970 valid clock-aligned trials over two sessions and 99 shared stimulus words. The descriptive pooled correlation between sessions across 471 matched participant–word pairs was weak ($r = 0.24$); it is not an estimate of participant-level or map reliability. Mean cross-participant latency correlation was small ($\bar{r} = 0.065$; exploratory two-sided permutation $p = 0.038$), while the first principal-component variance share (27%) was close to its shuffle-null mean (approximately 26%; $p = 0.21$). An in-sample rank-[3, 6, 3] tensor approximation described 75.9% of variance. The available records could be processed into descriptive profiles, but the weak pooled repeatability, small selected sample, and incomplete sensor provenance limit their interpretation. No independent criterion for psychological meaning was measured. This feasibility study does not validate a psychological construct or a clinically useful measurement in this sample.

Keywords: word association; response latency; skin potential; test–retest reliability; feasibility study

1 Introduction

Word-association latency and autonomic activity can vary with the stimulus and the participant (Jung, 1910). Graph and tensor methods can summarize these multivariate measurements (Kolda and Bader, 2009), but a constructed geometry has no psychological meaning without reliability and independent validation.

This pilot asks whether existing word-association events and skin-potential recordings can be aligned to form participant-specific descriptive profiles, and how repeatable the response latency is across the two sessions. We report the available sample, missing records, signal-processing choices, and exploratory cross-participant summaries so that the limits of the resulting profiles can be assessed. No independent criterion was collected to validate what the profiles represent.

2 Methods

2.1 Ethics and consent

Project materials report approval by Niigata University (approval 2024-0269; March 1, 2025). The study interface required acknowledgment of the research purpose, voluntary participation, the right to withdraw without disadvantage, and audio/video recording before registration.

2.2 Participants and analytic set

Eleven participant directories were present. Six could not enter the complete-record analysis: three continuous-recording files were unavailable git-annex pointers, two directories contained no physiological CSV, and one recording did not overlap the task-event clock. Five participants contributed analyzable synchronized records. This was an availability-based analytic subset, not a random or prospectively selected sample. No demographic or clinical inference is attempted.

2.3 Task and signal extraction

Participants completed two presentations of an approximately 100-word stimulus list. The present analysis uses stimulus identity, event timing, and continuous skin-potential signals; semantic content of spoken association responses is not used in the manifold.

Continuous Mod-002 CSV recordings were sampled at 1 Hz. Response latency was the interval from `response_window_opened` to the first `speech_detected` event; values had to satisfy $0 < \text{latency} < 12$ s. For each channel, skin-potential change ΔSP was the maximum absolute deviation during the response window from the mean of the 3 s preceding stimulus display. A trial was retained only when event times fell inside the recording and all four channel windows were available. This yielded 970 valid trials (198, 181, 195, 199, and 197 by participant).

Channel 2 is the primary skin-potential channel in the current analysis code; Channel 1 is retained as an additional tensor feature. This choice was not preregistered, and electrode-placement metadata sufficient to give the channels a stable physiological interpretation were unavailable.

2.4 Participant-specific response spaces

Within each participant, trials were averaged by stimulus word. The three features were latency, ΔSP_{Ch1} , and ΔSP_{Ch2} . For visualization we define a descriptive response-cost score

$$C(w) = z\{\text{latency}(w)\} + z\{\Delta SP_{Ch2}(w)\}. \quad (1)$$

This dimensionless index combines latency and skin-potential change for display. It is not a measure of self-inclusion, physiological energy expenditure, or information-thermodynamic work.

After feature standardization, squared Euclidean distances generated the radial-basis affinity

$$W_{ij} = \exp\{-\|x_i - x_j\|^2/\sigma^2\}, \quad (2)$$

where σ was the median nonzero pairwise distance. Coordinates were the nontrivial low-order eigenvectors of the symmetric normalized graph Laplacian $L_{sym} = I - D^{-1/2}WD^{-1/2}$. This is a graph-spectral response embedding, not Fisher information geometry.

2.5 Cohort tensor and exploratory statistics

Participant-by-word means were assembled as $\mathcal{T} \in \mathbb{R}^{P \times W \times F}$ with $P = 5$, $W = 99$, and $F = 3$. Missing cells were imputed first by the word–feature mean and then by the feature mean. Each feature was standardized across the participant–word plane. A truncated HOSVD/Tucker approximation (Kolda and Bader, 2009) used ranks $[3, 6, 3]$. The graph spectrum, participation ratio, and reconstructed variance are descriptive summaries whose ranks were not selected on an independent sample.

A descriptive pooled test–retest statistic was the Pearson correlation of session-1 and session-2 latency over available matched participant–word pairs. Because pairs are clustered within five participants and words, this pooled statistic is not an independent-observation estimate of population reliability or participant-specific map stability. Cross-participant structure was evaluated from the participant-by-word latency matrix. The mean off-diagonal Pearson correlation and first-principal-component variance share were compared with null distributions from 2,000 within-participant word permutations using seed 7. The correlation test was two-sided. These analyses were exploratory, were not preregistered, and no multiplicity-adjusted confirmatory claim is made.

3 Results

3.1 Trial-level response landscape

All 970 valid trials are shown in Fig. 1; no percentile trimming is used in the revised display. Because color is computed from the plotted axes, separation of low and high C is expected by construction. It demonstrates measurement coverage, not latent classes. The descriptive pooled session-1/session-2 latency correlation was weak ($r = 0.24$, 471 matched participant–word observations). Participant-level reliability estimates and confidence intervals remain to

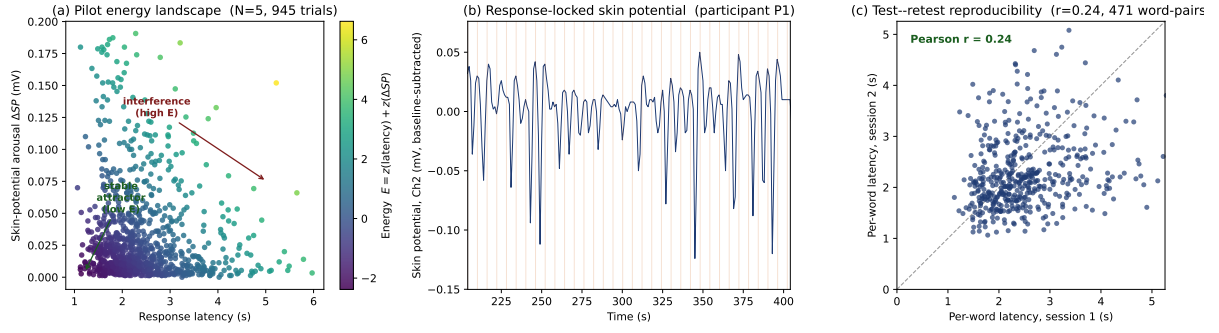


Figure 1: **Pilot measurements.** (a) Response latency and Channel-2 skin-potential change for 970 valid trials, colored by the descriptive response-cost score C . (b) Example continuous skin-potential trace and task events. (c) Session-1 versus session-2 latency for matched participant–word observations.

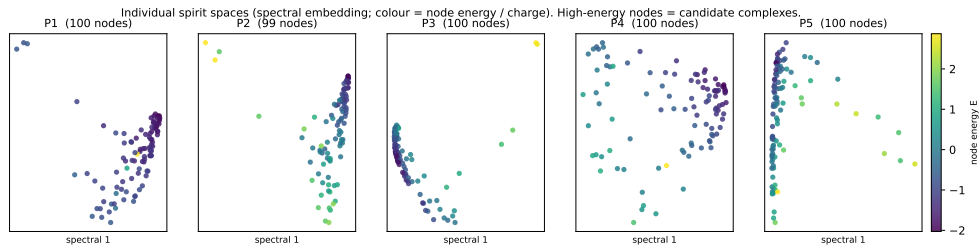


Figure 2: **Participant-specific response spaces.** Each panel is a graph-spectral embedding of one participant’s word nodes from latency and two skin-potential features; color is C . Axes have arbitrary sign and orientation and are not directly comparable without alignment.

be calculated from a disclosure-reviewed derived table.

3.2 Participant-specific embeddings

The same algorithm produced a two-dimensional response embedding for each participant (Fig. 2). Visual differences are descriptive. No independently assessed personal-salience or complex criterion, or validated classifier, is available in this pilot.

3.3 Cross-participant latency and tensor summaries

Mean off-diagonal cross-participant latency correlation was $\bar{r} = 0.065$ and exceeded the absolute shuffled statistic in the two-sided permutation test ($p = 0.038$). The first principal component explained 27%, close to the approximately 26% shuffle-null mean ($p = 0.21$; Fig. 3). This small exploratory association can reflect common lexical or task effects. The principal-component result does not support a large common component; no Jungian construct was tested.

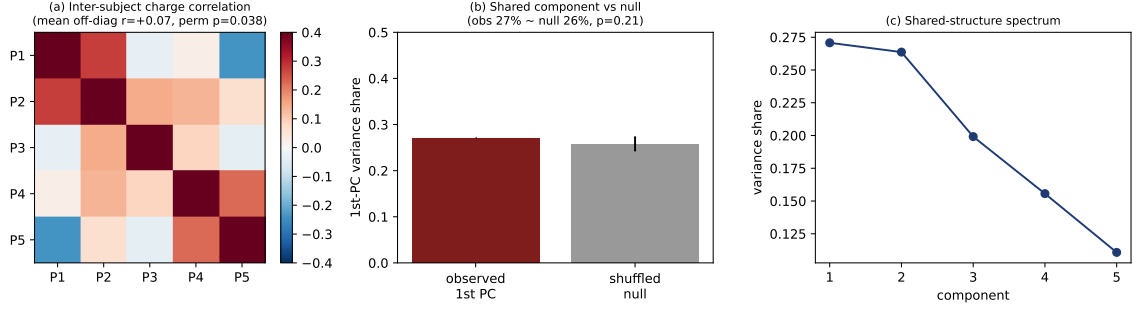


Figure 3: **Exploratory cross-participant latency analysis.** (a) Pairwise correlations and their mean, with a two-sided within-participant permutation test. (b) Observed first-PC variance share versus the shuffle null. (c) Latency variance spectrum. These panels analyze latency, not a physiological “charge.”

The tensor had shape $5 \times 99 \times 3$. The median-distance kernel scale was $\sigma = 4.2109$, the graph eigengap was 0.1948, and the spectrum’s participation ratio was 10.89. Rank- $[3, 6, 3]$ HOSVD reconstructed 75.9% of variance. With only five participants and three features, these values partly reflect dimensional constraints and do not establish a population manifold or a unique “collective coordinate.”

4 Discussion

The analysis shows that the available event and signal records can be transformed into word-linked descriptive profiles. The principal measurement issue is weak pooled latency repeatability, which limits the interpretation of any derived map. The five-person availability-based subset also creates selection risk. These results justify improving acquisition and evaluating reliability in a separately designed study; they do not establish the validity of a psychological measure.

The pooled latency correlation is not an estimate of participant-level or map reliability. The small cross-participant correlation is exploratory and may reflect shared words or task structure; the first-PC comparison did not distinguish the observed result from the shuffle null. The in-sample tensor reconstruction likewise does not show predictive validity. The response-cost score and graph coordinates are descriptive transformations of the recorded features.

The next empirical study should first address the missing acquisition records, channel

metadata, participant-level repeatability, and independent validation criteria. Its protocol and sample-size rationale require separate approval before data collection.

5 Limitations

The analytic sample was five of eleven participants and depended on local data availability. Missing raw annex objects and absent or misaligned recordings may introduce selection bias. Sampling at 1 Hz limits temporal interpretation. Channel choice was not preregistered, sensor metadata were incomplete, and no negative-control stimulus set or independent validation cohort was available. Semantic association responses and rubber-hand-illusion data were not analyzed. Statistics were exploratory and no adjustment for multiple analytic choices was applied. Participant-level map repeatability and criterion validity were not measured. Age, gender and other sample characteristics have not yet been verified against source records and cannot be reported in this draft. No prospective sample-size calculation was performed for this availability-based pilot. The current repository is not publicly accessible and the deidentified derived dataset is not yet deposited in a durable public archive.

6 Data and code availability

The analysis code, event-processing logic, and manuscript sources are maintained in the project repository. At the time of manuscript preparation the repository is private and participant-level recordings are not publicly released. A deidentified derived table, versioned code snapshot, environment specification, and checksums have not yet been deposited in a durable archive; the present analysis should therefore not be described as independently reproducible. Access to underlying research data is subject to ethics approval, consent, and an appropriate data-use process; inquiries may be directed to the corresponding author.

7 Conclusion

This five-participant pilot shows that the available word-association and skin-potential records can be transformed into descriptive response profiles. The weak pooled latency correlation, missing sensor provenance, and lack of independent validation prevent a reliability or validity claim for the derived profiles. A larger, prospectively specified reliability study is needed before these profiles can support psychological interpretation.

References

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